



Physical Buttons For Tactile Future

Physical buttons may seem outdated in the touchscreen era, but the University of Illinois Urbana (IL, United States)-Champaign team is revolutionising them with



HaptiCoil buttons sense touches and provide haptic feedback (Credit: Interactive Display Lab)

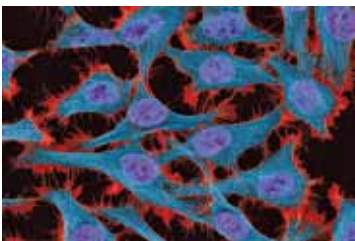
HaptiCoil. This system transforms traditional buttons into dynamic, tactile interfaces to rapidly prototype customisable, soft haptic buttons. HaptiCoil delivers a wide range of sensations via a waterproof planar micro-speaker and a hydraulic mechanism using tap water. It detects button presses by measuring changes in the speaker coil's self-inductance. Compact, durable, and versatile, HaptiCoil is ideal for industries like spatial computing and digital inking, offering new possibilities in tactile interaction.

Dual-Scalable Processor Solves Optimisation Challenges

The Team at the Tokyo University of Science (Shinjuku, Japan) has developed a dual scalable annealing processing system (DSAPS) to tackle combinatorial optimisation problems (COPs) more accurately and efficiently. COPs, critical in fields like scheduling and drug design, are challenging for traditional computing. DSAPS overcomes limitations in previous annealing processors by offering scalable structures for both capacity and precision. It utilises high capacity and precision energy computation blocks, enabling advanced configurations, such as 2048 spins with 10-bit interactions, delivering superior performance compared to traditional ASIC designs.

Precise Gene Therapy Control

Gene therapy shows promise for treating diseases caused by defective genes, like haemophilia and sickle cell anaemia, but challenges in controlling gene expression have



Precise gene therapy control (Credit: <https://news.mit.edu>)

limited FDA-approved treatments. MIT's (USA) new approach solves this with a control circuit ensuring precise regulation. This system, called ComMAND, uses an incoherent feedforward loop (IFFL) to finely

tune gene expression. It can regulate gene expression for conditions like fragile X syndrome and Friedreich's ataxia, offering a more reliable and manufacturable solution. MIT's innovation could improve treatment for rare genetic disorders like Rett syndrome and muscular dystrophy.

Flying Squirrel Drone Shows Agility

Engineers at Pohang University of Science and Technology, (Pohang, South Korea) alongside South Korea's Agency for Defense Development, have developed a drone



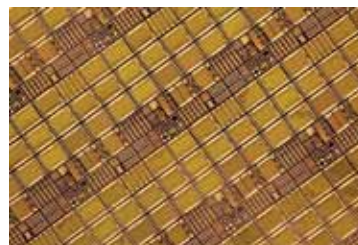
Flying squirrel drone shows agility (Credit: <https://newatlas.com>)

inspired by the flying squirrel. It features foldable, silicone-based wings that enhance manoeuvrability by creating aerodynamic drag, allowing sharp turns and rapid stops. Using a neu-

ral network, the drone's control system coordinates wing deployment and motor thrust in real time. With onboard computing, this lightweight, energy-efficient drone promises applications in remote monitoring, search and rescue, defence, and cinematography.

Power Electronics Transform Energy And Mobility Sectors

The Fraunhofer Institute for Applied Solid State Physics IAF (Breisgau, Germany) is presenting advancements in power electronics with gallium nitride (GaN) technology,



Monolithic bidirectional 1200V GaN switches (MBDS) with integrated free-wheeling diodes, made at Fraunhofer IAF in a multi-project wafer run using GaN-on-insulator technology (Courtesy of Fraunhofer IAF)

aiming to accelerate the energy transition and enhance Europe's economic resilience. Their key innovation is a 1200V monolithic bidirectional switch (MBDS), ideal for grid-connected converters and electric vehicle systems. Developed in the GaN4E-moBiL project, it integrates

free-wheeling diodes and uses GaN-on-insulator technology with substrates like silicon carbide and sapphire. This reduces conduction losses, saves chip space, and improves charging speed, efficiency, and power density in next-gen electric vehicles and renewable energy systems.